

MTH203: Multivariate Calculus

Monsoon 2025

Name: _____

Roll No.: _____

Tutorial Quiz 7

Question:

1. Write $\mathbf{a}$ in the form $a_T\mathbf{T} + a_N\mathbf{N}$ without finding $\mathbf{T}$ and $\mathbf{N}$ for the motion

$$\mathbf{r}(t) = (a \cos t)\mathbf{i} + (a \sin t)\mathbf{j} + bt\mathbf{k} \quad (1)$$

2. A wire of density $\delta(x, y, z) = 15\sqrt{y+2}$ lies along the curve

$$\mathbf{r}(t) = (t^2 - 1)\mathbf{j} + 2t\mathbf{k}, \quad -1 \leq t \leq 1$$

Find its center of mass. (2)

Please note: There is no fractional marking. Only integer marking.